Macroeconomic Asset Divergence Model

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Objective

- We would like to construct macroeconomic models for each of the 11 *Global Industry Classification Standard* (GICS) sectors.
- Given those models, we watch for divergence of actual sector returns with those predicted by our model.
- We will use this as a trading signal to long or short a sector in a constructed portfolio.

Quant Insight Methodology

- For each sector, we would like to find the most important macroeconomic factors in predicting returns.
- We would like to establish a universal factor basis that explains each sector of the economy well.
- *Envision by Quant Insight* [1] starts with a large number of macroeconomic factors that potentially drive a given sector.

Quant Insight Methodology

- However, many of these factors are highly collinear.
- To circumvent this, they leverage PCA which converts these potentially correlated variables into a set of linearly uncorrelated orthogonal principal components.
- Model each sector on principal components and define the aforementioned trading signals.

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Forward Stepwise Selection

- Suppose we have a data matrix $X \in \mathbb{R}^{m \times n}$ of weekly factor returns.
- We split the data into an in-sample period $X_{\text{in-sample}} \in \mathbb{R}^{k \times n}$ (2010-2019) and out-of-sample period $X_{\text{out-of-sample}} \in \mathbb{R}^{m-k \times n}$ (2019-2023).
- For each sector, we want to predict index returns y_t for $1 \leq t \leq k$ using coincident factor return data.

Forward Stepwise Selection

- Let $\mathcal{M}_0$ be the null linear model with no predictors, i.e. $y_t = \beta_0$.
- For $i = 0, \ldots, p-1$, we consider all $p - i$ models that augment predictors in $\mathcal{M}_i$, defined as:

$$y_t = \beta_0 + \beta_1 x_{t,1} + \cdots + \beta_i x_{t,i}$$

  with one new predictor.
- We select the best model, which has the smallest RSS (or via statistical significance with a t-test), when measure on in-sample data $X_{\text{in-sample}}$:

$$\begin{aligned} \hat{y}_t &= \hat{\beta}_0^{\text{OLS}} + \hat{\beta}_1^{\text{OLS}} x_{t,1} + \cdots + \hat{\beta}_i^{\text{OLS}} x_{t,i} \\ RSS(\hat{\beta}^{\text{OLS}}) &= \sum_{t=1}^{k} (y_t - \hat{y}_t)^2 \end{aligned} \tag{1}$$

- We get a greedily selected model $\mathcal{M}_p$ for $p < n$, instead of best subset selection.

Example Sector: S5MATR Index

- Selected factors/coefficients for materials sector:
  - BP1 Curncy: 0.16 - British Pound
  - GC1 Comdty: 0.14 - Gold
  - MXWD Index: 0.05 - MSCI World Index
  - SP5HBI Index: 0.43 - S&P 500 High Beta Index
  - SVX Index: 0.42 - S&P 500 Value Index
- In-sample R^2 from 2010-2019: 0.831.

Dimensionality Reduction

- We did forward stepwise selection for each of the 11 GICS sectors with $p = 5$.
- Next, we look at the superset of these predictors.
- In a similar vein to the Quant Insight methodology [1], we perform a principal components analysis (PCA) with three principal components.
- These principal components will explain the most amount of variance of the data in mutually orthogonal directions.

Singular Value Decomposition

- Consider a data matrix $X \in \mathbb{R}^{n \times p}$.
- The SVD of a $n \times p$ matrix X has the form [2]:

$$X = UDV^\top \tag{2}$$

  for U and V are $n \times p$ and $p \times p$ orthogonal matrices, whereby $span(\mathbf{u}_1, \ldots, \mathbf{u}_p) = C(X)$ and $span(\mathbf{v}_1, \ldots, \mathbf{v}_p) = C(X^\top)$.
- D is a $p \times p$ diagonal matrix with diagonal entries called the *singular values* of X.

SVD for Sample Covariance

- The sample covariance is $\Sigma = \frac{1}{n} X^\top X$. Apply SVD:

$$X^\top X = V D^2 V^\top, \tag{3}$$

  the eigen decomposition (diagonalization) of $X^\top X$.
- Eigenvectors $\mathbf{v}_j$ are the principal component directions of X.
- Principal component direction (eigenvector) $\mathbf{v}_1$ has property $\mathbf{z}_1 = X\mathbf{v}_1$ has largest sample variance among linear combinations of columns of X.
- Sample variance:

$$\mathrm{Var}(\mathbf{z}_1) = \mathrm{Var}(X\mathbf{v}_1) = \frac{d_1^2}{n}. \tag{4}$$

Principal Components Analysis

- $\mathbf{z}_1 = X\mathbf{v}_1 = \mathbf{u}_1 d_1$ is the first principal component of X.
- More generally, columns of UD are principal components.
- Subsequent principal components $\mathbf{z}_j$ have maximum variance d_j^2/n subject to being orthogonal to those preceeding it.
- To reduce dimensionality of the data from p predictors to k, we select the k first columns of U and the $k \times k$ upper-left part of D. The product $U_k D_k$ is a $n \times k$ matrix of the first k PCs.
- Multiplying the first k PCs by the principal directions $V_k^\top$, we get a $n \times k$ matrix $X_k = U_k D_k V_k^\top$.

PCA $k = 3$ Visualization

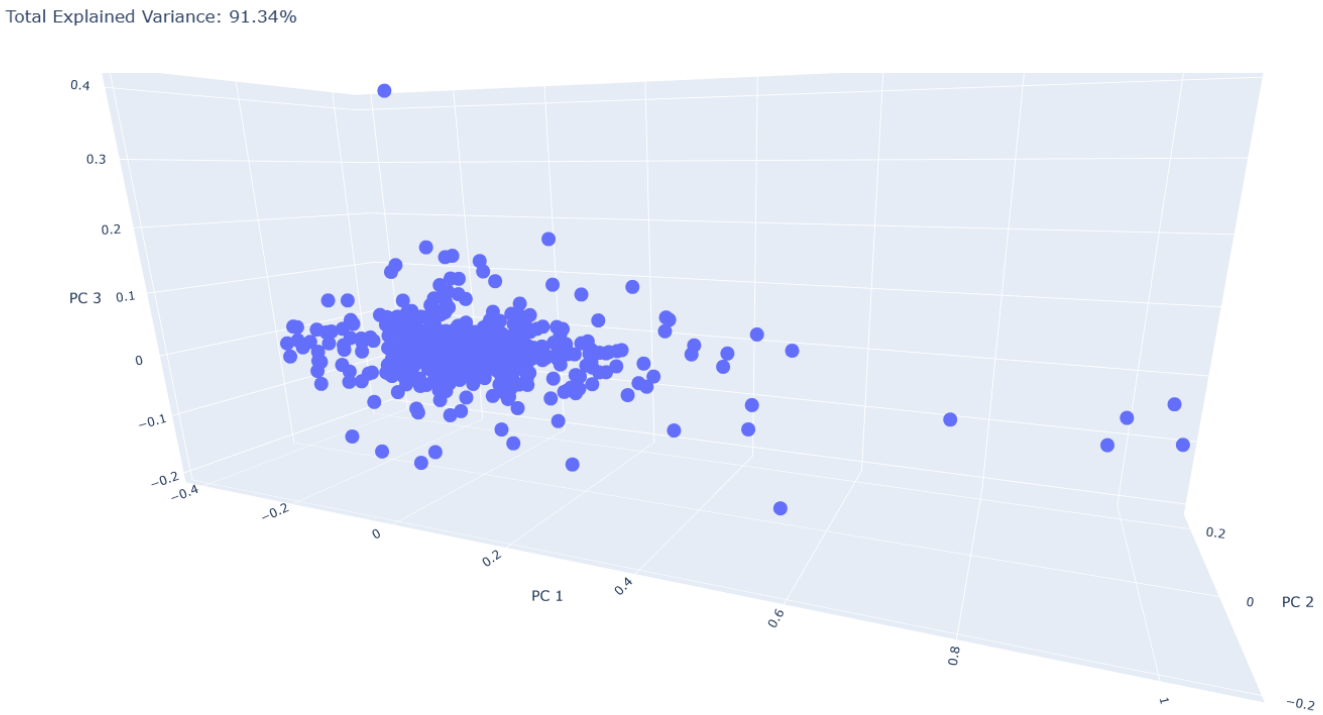


Figure: $k = 3$

Cumulative Variance Explained

We run PCA on the normalized matrix of superset factors, obtaining a reduced $X_k \in \mathbb{R}^{n \times k}$ for $k = 5$, with explained variance of 91.34%.

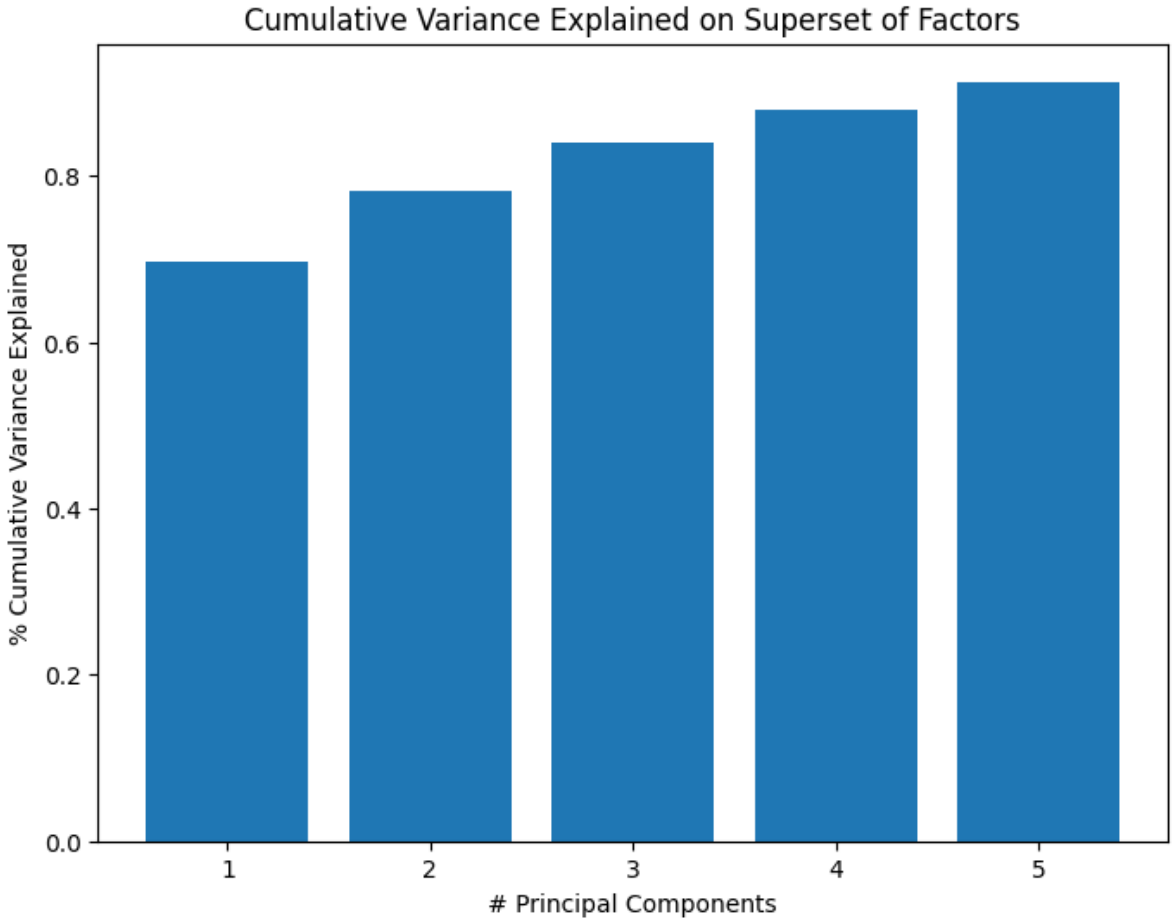


Figure: Cumulative Variance Explained

Meaning of PCs

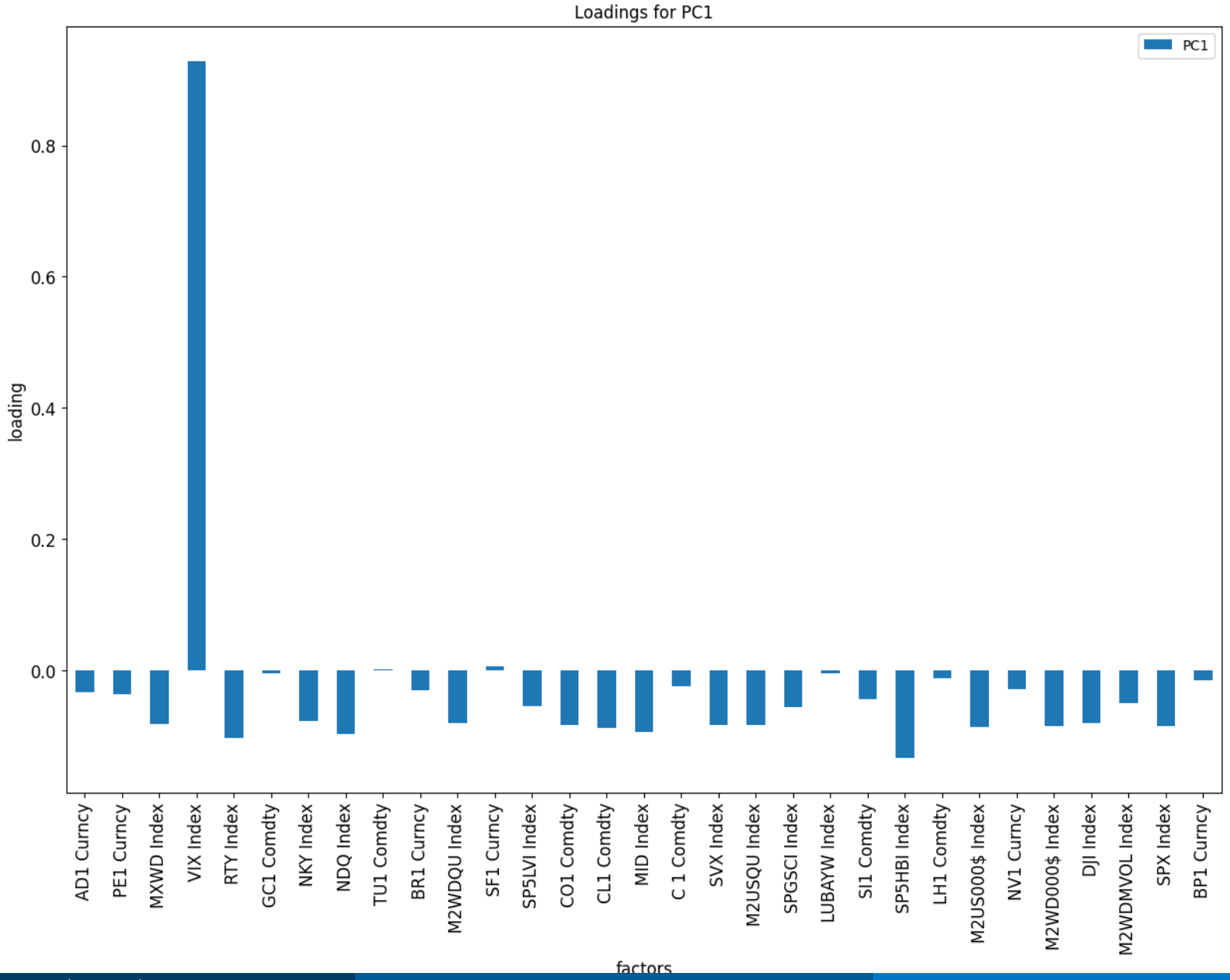
- Recall that the sample covariance is:

$$\begin{aligned} \frac{1}{n} X^\top X &= V D U^\top U D V^\top \\ &= V \frac{D^2}{n} V^\top \\ &= V \Lambda V^\top \end{aligned} \tag{5}$$

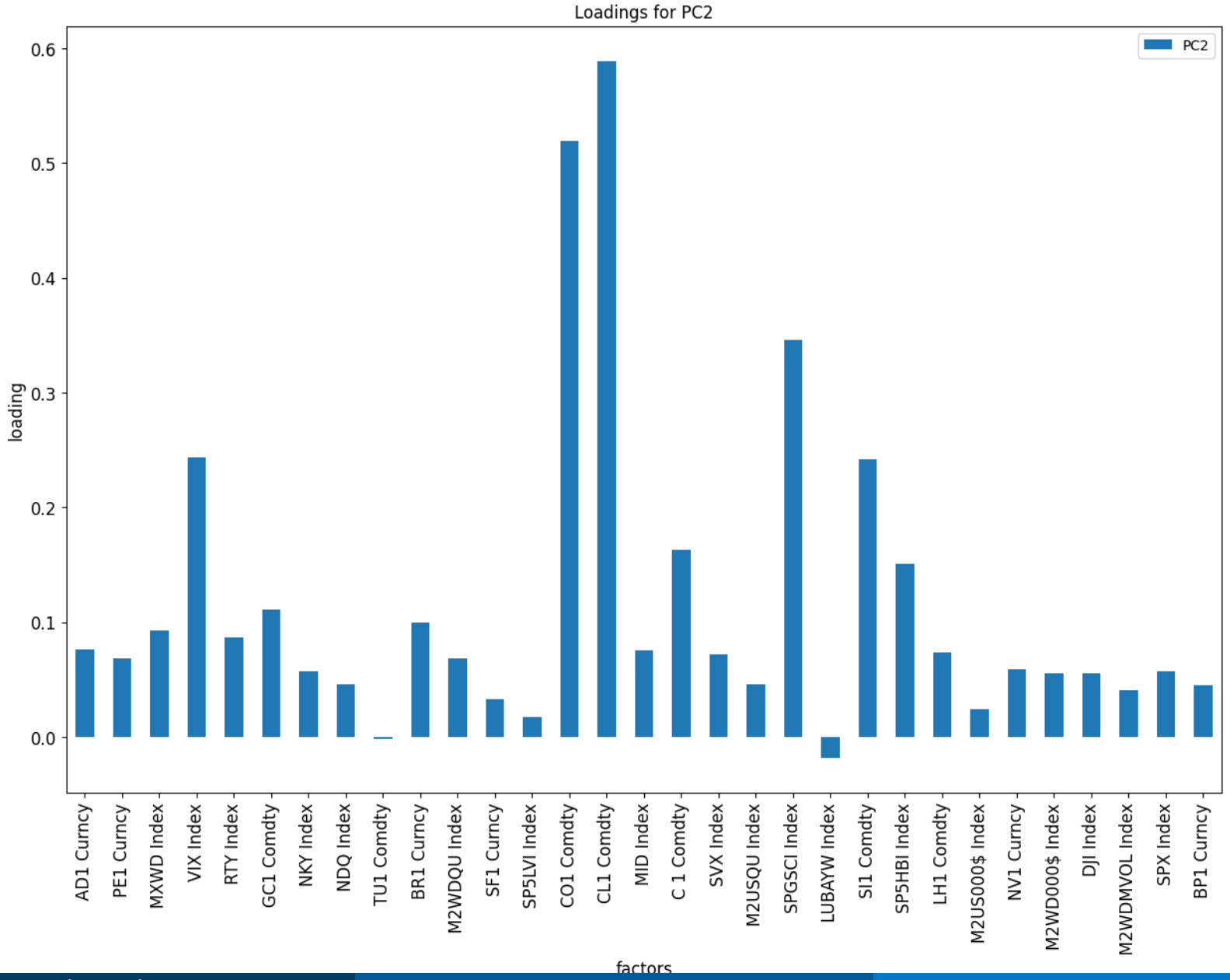
- Hence, the eigenvalues of the covariance matrix are given by $\lambda_i = d_i^2/n$; the eigenvalues show the variances of the respective PCs.
- Loadings are the correlations between principal components and factors, given by the columns of

$$V\sqrt{\Lambda} = VD/\sqrt{n} \tag{6}$$

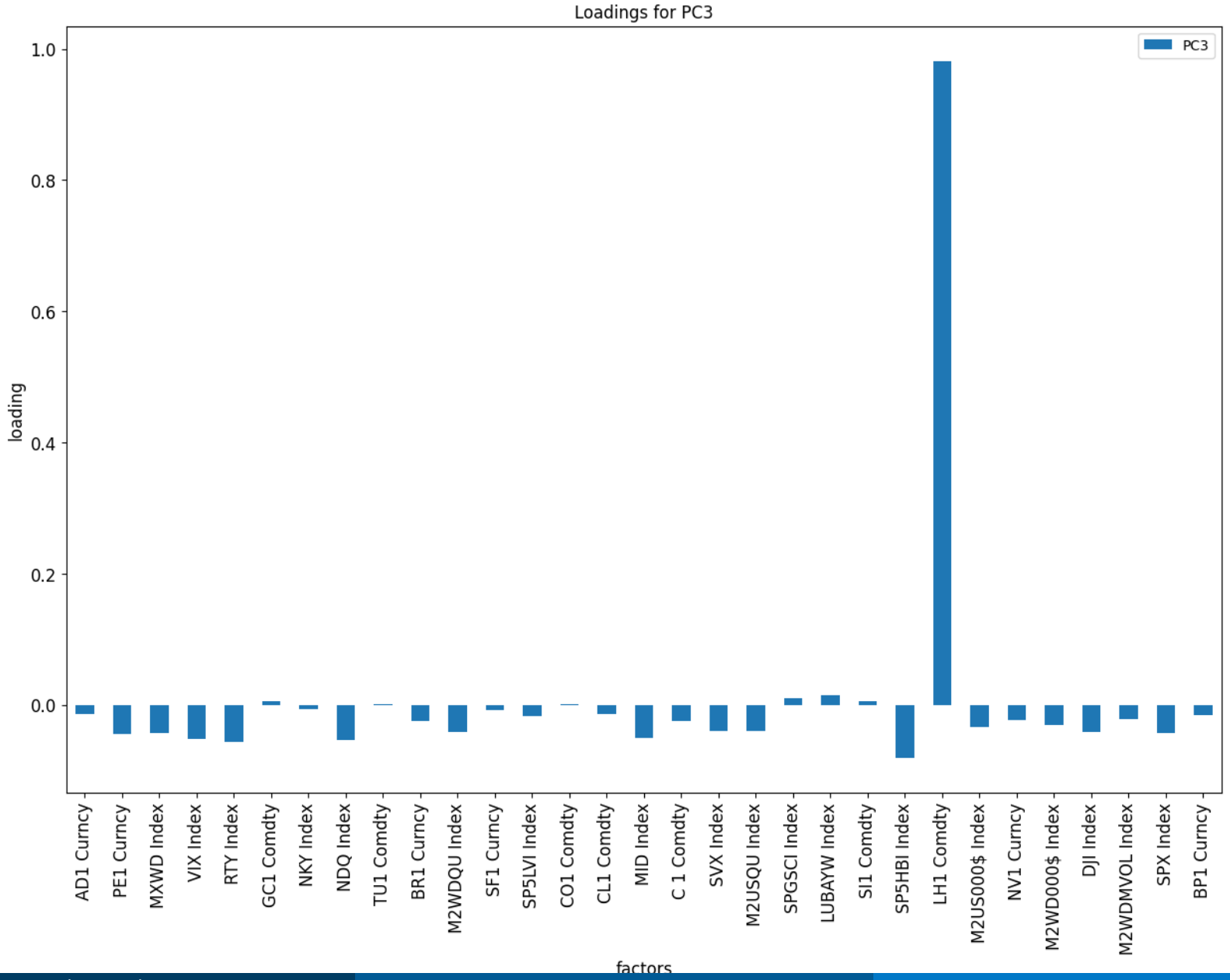
Loadings of PC1 onto superset of factors



Loadings of PC2 onto superset of factors



Loadings of PC3 onto superset of factors



Meaning of PCs

- We find that the principal components with the highest loadings (in absolute value) are given as follows:
- PC1: 'VIX Index' - Volatility
- PC2: 'CL1 Comdty' - Crude Oil
- PC3: 'LH1 Comdty' - Lean Hogs
- PC4: 'SI1 Comdty' - 10 Year Treasury Note
- PC5: 'SI1 Comdty' - 10 Year Treasury Note

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Coincident Factor Model & Assumptions

- The principal components are used to perform a time-series principal component regression (PCR) against the sector index.
- Suppose we would like to predict y_t, i.e. the current week's returns. We allow ourselves to use coincident returns of other relevant factors over the past $N = 469$ weeks $\mathbf{x}^{(t)}, ..., \mathbf{x}^{(t-N+1)}$ from 2010 to 2019:

$$X_{\text{in-sample}} = \begin{bmatrix} x_1^{(t)} & x_2^{(t)} & x_3^{(t)} \\ \vdots & \vdots & \vdots \\ x_1^{(t-N+1)} & x_2^{(t-N+1)} & x_3^{(t-N+1)} \end{bmatrix}, y_{\text{in-sample}} = \begin{bmatrix} y_t \\ \vdots \\ y_{t-N+1} \end{bmatrix}. \tag{7}$$

Fitting a Model

We use an exponentially weighted least squares, whereby we estimate $\hat{\beta}$ through the following objective:

$$\hat{\beta} = \arg\min_{\beta} \sum_{i=1}^{N} w_i (y_{t-i+1} - \hat{y}_{t-i+1})^2, \tag{8}$$

and $w_1 = 1, w_2 = \lambda, \ldots, w_N = \lambda^{N-1}$. In matrix notation:

$$(X^\top W X)\hat{\beta} = X^\top W y \tag{9}$$

where

$$W = \begin{bmatrix} 1 & & & \\ & \lambda & & \\ & & \ddots & \\ & & & \lambda^{N-1} \end{bmatrix}. \tag{10}$$

Fitting a Model

- The normal equations can be written in the usual OLS notation as:

$$(X'^\top X')\hat{\beta} = X'^\top y \tag{11}$$

where $X' = \text{diag}(w)X$ and $y' = \text{diag}(w)y$. Suppose that we have pre-processed the data using weights, for simplicity of notation.

- Hence, we estimate parameters as:

$$\hat{\beta} = (X_{\text{in-sample}}^\top X_{\text{in-sample}})^{-1} X_{\text{in-sample}}^\top y_{\text{in-sample}}. \tag{12}$$

- We measure goodness of fit with R^2.

Fitting a Model

Using this estimate and the returns from week t, we can now predict future returns for week $t+1$ given factor returns in the same week:

$$\begin{aligned}\hat{y}_{t+1} &= \begin{bmatrix} x_1^{(t+1)} & x_2^{(t+1)} & x_3^{(t+1)} \end{bmatrix} \begin{bmatrix} \hat{\beta}_1 \\ \vdots \\ \hat{\beta}_p \end{bmatrix} \\ &= X_{\text{new}}\hat{\beta}.\end{aligned} \tag{13}$$

Materials Sector Example

- Goodness of fit using top-5 sector-level factors vs PCA in-sample from 2010-2019: $R^2 = 0.831$ vs $R^2 = 0.773$.
- Goodness of fit using top-5 sector-level factors vs PCA out-of sample: $R^2 = 0.847$ vs $R^2 = 0.829$.
- We are using a universal PCA to get principal components and, hence, identify the main macroeconomic drivers in sectors.
- Note, there might be idiosyncratic drivers specific to sectors that we could substitute in this basis if including it augments R^2 significantly on a sector basis.
- PCA yields a basis that is mutually orthogonal, i.e. we eliminate collinearity in factors to get the main drivers from the supserset.

Defining Spread

- We define a spread:

$$s_t = y_t - \hat{y}_t, \tag{14}$$

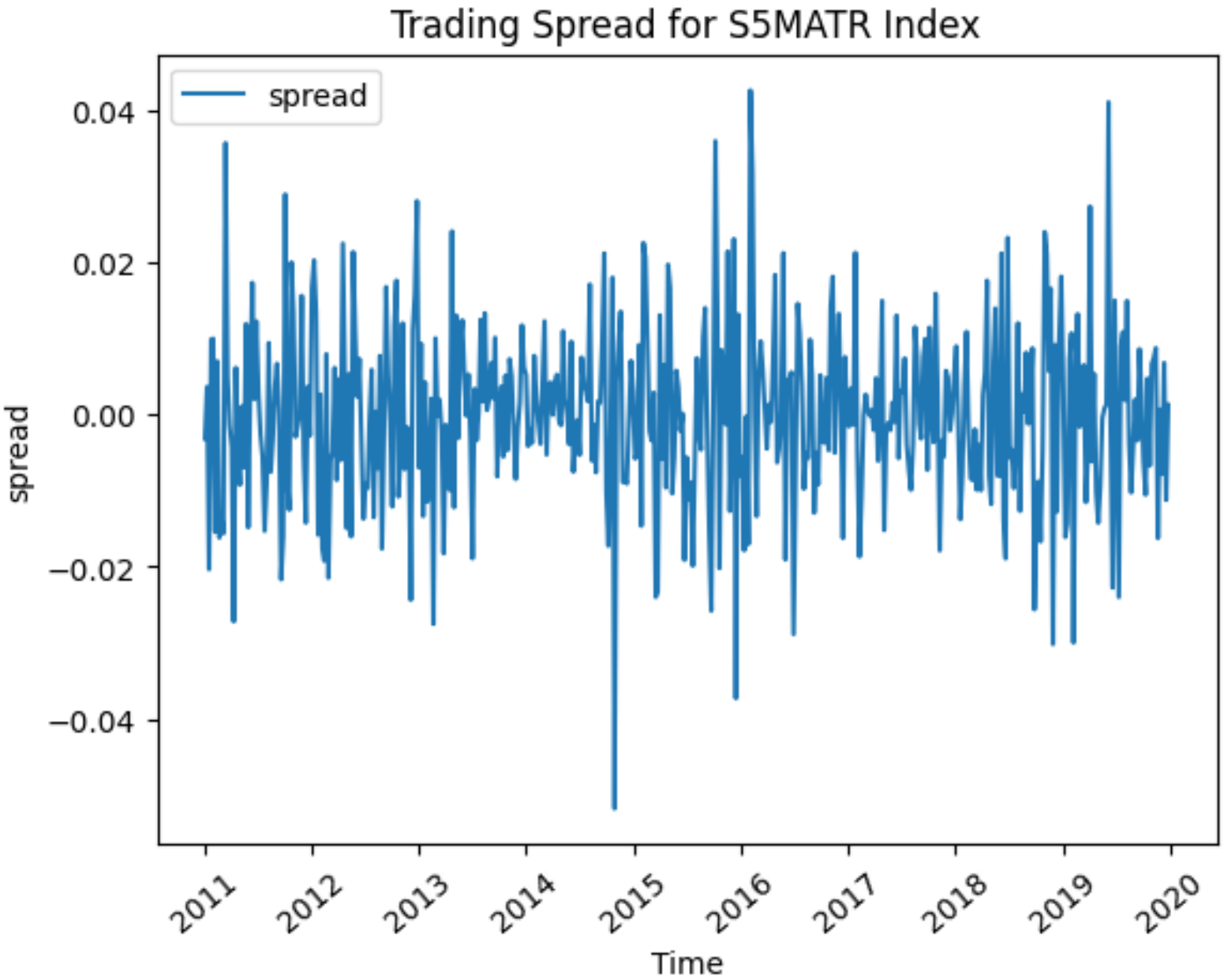
namely, how much actual returns differs from our macroeconomic model.

- Indeed, since $s_t = \epsilon_t \sim \mathcal{N}(0, \sigma^2)$, we hypothesize that the process is stationary.
- That is, one can model s_t as a stationary AR(1) process:

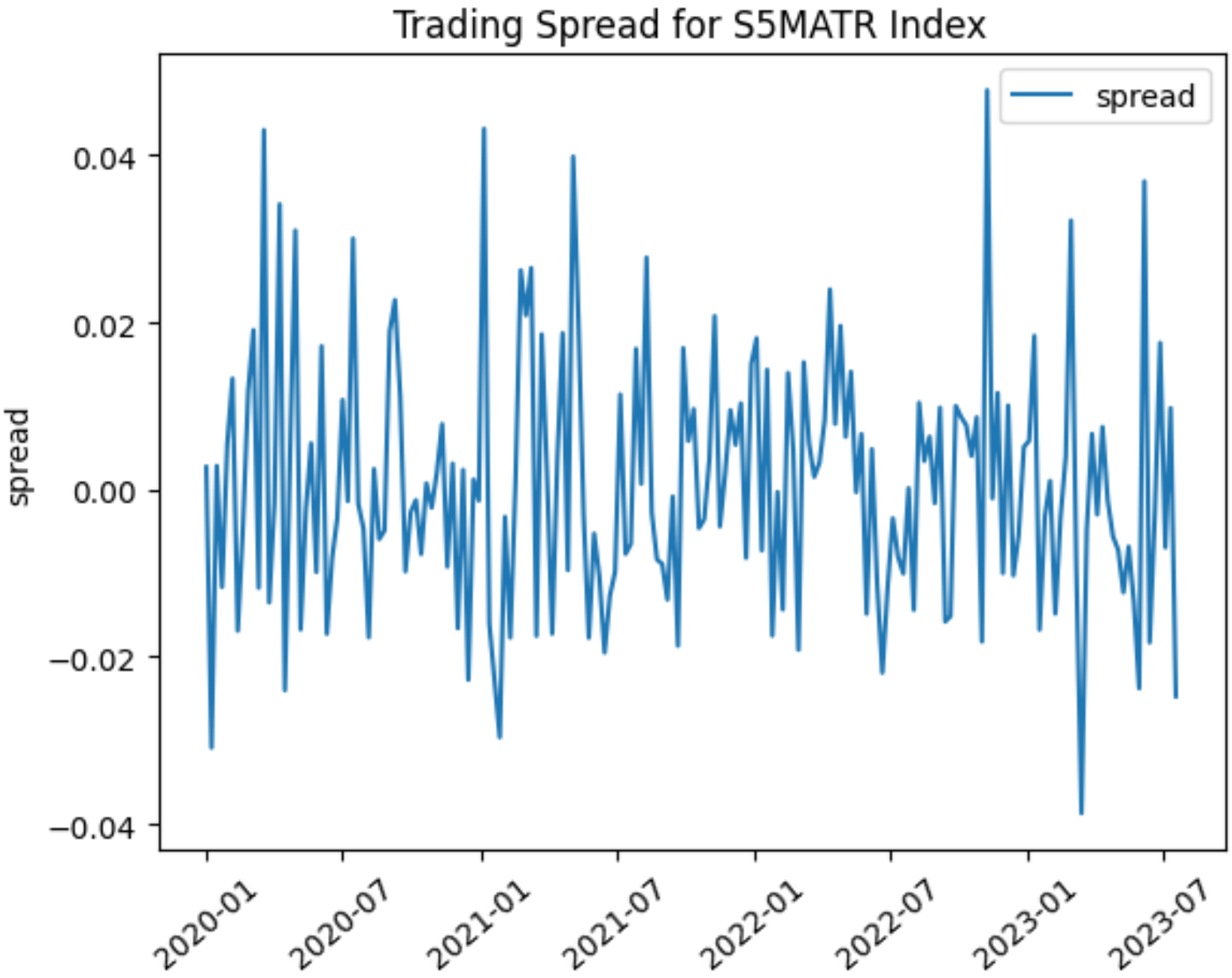
$$s_t = \rho s_{t-1} + \varepsilon_t \tag{15}$$

for $|\rho| < 1$ and $\varepsilon_t \overset{i.i.d.}{\sim} (0, \sigma^2)$.

In-Sample Spread



Out-of-Sample Spread



Stationarity

Definition

A stochastic process $(X_t : t \in T)$ is **stationary** if for all $t_1, \ldots, t_n \in T$ and h such that $t_1 + h, \ldots, t_n + h \in T$, the finite-dimensional distributions of the process are invariant under time shift:

$$(X_{t_1}, \ldots, X_{t_n}) \overset{\mathrm{D}}{=} (X_{t_1+h}, \ldots, X_{t_n+h}) \tag{16}$$

Weak Stationarity

Definition

A stochastic process X_t is **weakly stationary** if:

- $\mathbb{E}[X_t] = \mu < +\infty$
- $\mathrm{Var}[X_t] = \sigma^2 < +\infty$
- $Cov(X_t, X_{t+h}) = \mathbb{E}[(X_t - \mu)(X_{t+h} - \mu)] = f(h)$

Integrated Order of Process

Definition

Let L be the lag order such that

$$(1 - L)X_t = X_t - X_{t-1} = \Delta X_t. \tag{17}$$

A stochastic process X_t is of **integrated order** k if

$$(1 - L)^k X_t \tag{18}$$

is stationary.

Remark

It is generally assumed stock returns are $I(0)$ while stock prices are $I(1)$ [3].

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Testing for Stationarity

- To test for stationarity in our process, we use the Dickey-Fuller test.
- Suppose that X_t is a random walk with drift (s_t, in our case), given by

$$X_t = \alpha + \rho X_{t-1} + \epsilon_t, \tag{19}$$

where $\epsilon_t \overset{\text{i.i.d.}}{\sim} (0, \sigma^2)$

- Hence, consider the difference:

$$\begin{aligned} \Delta X_t &= X_t - X_{t-1} \\ &= \alpha + (\rho - 1)X_{t-1} + \epsilon_t \\ &= \alpha + \delta X_{t-1} + \epsilon_t. \end{aligned} \tag{20}$$

Hypothesis Test

- To test for stationarity, we use an autoregressive root test with the following hypotheses:

$$\begin{aligned} H_0 : \delta = 0 &\implies X_t \sim I(k) \text{ for } k > 0 \text{ (non-stationary)} \\ H_1 : \delta < 1 &\implies X_t \sim I(0) \text{ (stationary)} \end{aligned} \tag{21}$$

- We use a least squares estimate of $\hat{\delta}$ and, thus, the test statistic is:

$$t_{\delta=0} = \frac{\hat{\delta} - 0}{SE(\hat{\delta})} \tag{22}$$

where $SE(\hat{\delta})$ is the standard error estimate. This is a one-sided left tail test.

Refined Spread

- We tested the spread s_t and concluded that it is, indeed, I(0).
- However, we would like to get stronger, less noisy signals. One such way of doing this is to use a rolling sum over small size-N windows such that we retain I(0) properties, but still get trend-like properties.
- In particular,

$$\tilde{s}_t := \sum_{i=0}^{N} s_{t-i}. \tag{23}$$

- Other idea: use a cumulative spread with an exponentially-weighted moving average. However, it will be non-stationary due to changing stdev.
- Instead, we can run a Monte Carlo simulation of exponentially weighted moving averages, get an estimate for volatility, and standardize the cumulative spread.

Spread Property

Lemma

Let $s_t := y_t - \hat{y}_t \sim \mathcal{N}(0, \sigma^2)$ *be a stochastic process. Then, indeed,* $\tilde{s}_t = \sum_{i=0}^{N} s_{t-i}$ *is an* $I(1)$ *process.*

Proof of Property

Proof.

Suppose s_t is I(0). Now let $\tilde{s}_t = \sum_{i=0}^{N} s_{t-i}$. Then consider the first-time difference:

$$\begin{aligned} \Delta\tilde{s}_t &= \tilde{s}_t - \tilde{s}_{t-1} \\ &= \sum_{i=0}^{N} s_{t-i} - \sum_{i=0}^{N} s_{t-1-i} \\ &= s_t - s_{t-1-N} \end{aligned} \tag{24}$$

By construction, $s_t \overset{\text{i.i.d.}}{\sim} (0, \sigma^2)$. In particular, $s_t \sim \mathcal{N}(0, \sigma^2)$, then

$$s_t - s_{t-1-N} \sim \mathcal{N}(0, 2\sigma^2). \tag{25}$$

Hence, $\Delta\tilde{s}_t$ is $I(0)$ so $\tilde{s}_t$ is $I(1)$, as asserted. □

Weak-Stationarity of Spread

Lemma

Let $s_t := y_t - \hat{y}_t \sim \mathcal{N}(0, \sigma^2)$ *be a stochastic process. Then,* $\tilde{s}_t = \sum_{i=0}^{N} s_{t-i}$ *is weakly stationary.*

Proof of Weak-Stationarity

Proof.

Let $\tilde{s}_t = \sum_{i=0}^{N} s_{t-i}$ and recall, a priori, $s_t \sim \mathcal{N}(0, \sigma^2)$. Hence,

$$\tilde{s}_t = \sum_{i=0}^{N} s_{t-i} \sim \mathcal{N}(0, (N+1)\sigma^2). \tag{26}$$

Clearly, $\mathbb{E}[\tilde{s}_t] = 0$ and $\mathrm{Var}[\tilde{s}_t] = (N+1)\sigma^2$. Finally,

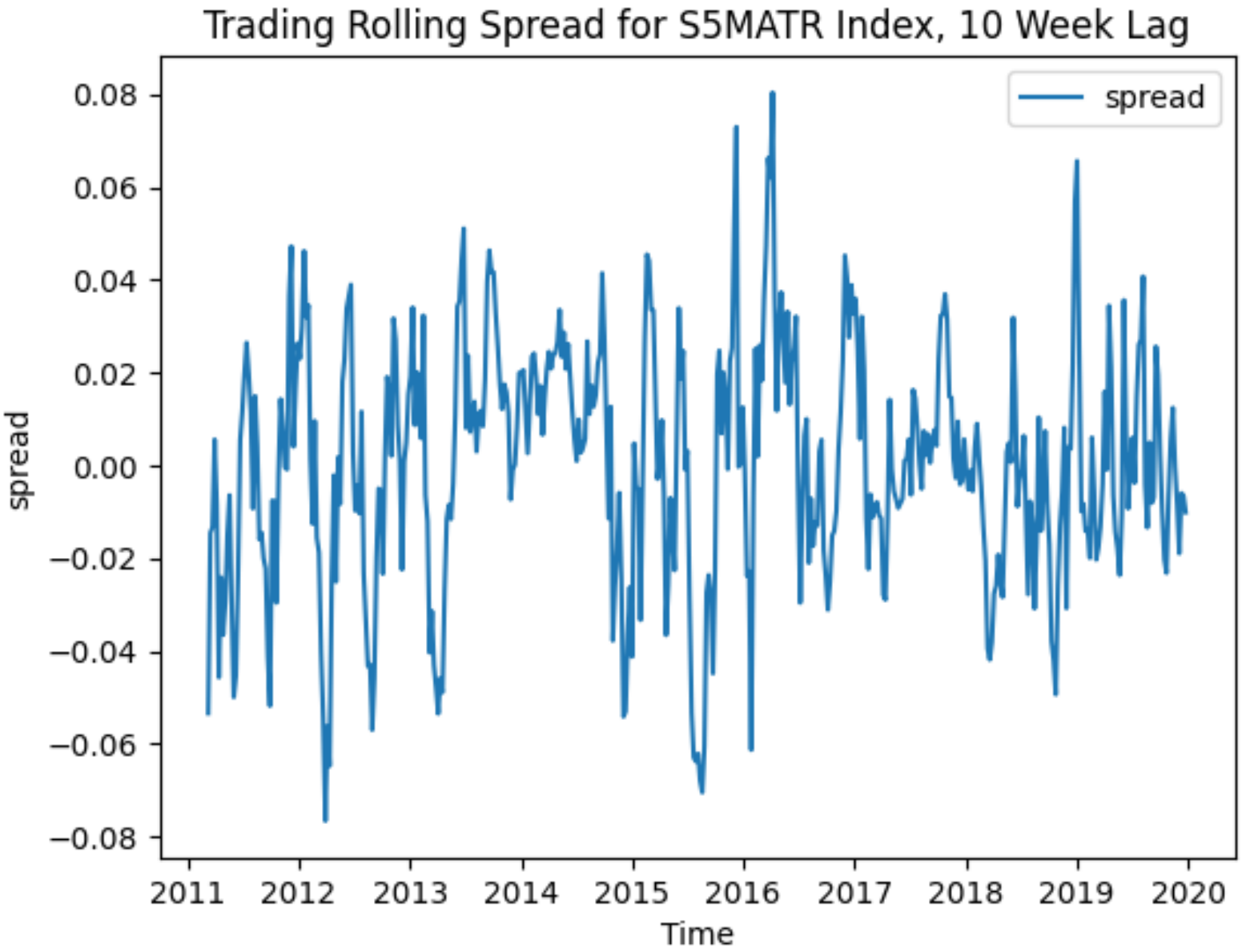
$$\begin{aligned} Cov(\tilde{s}_t, \tilde{s}_{t+h}) &= \mathbb{E}[(\tilde{s}_t - 0)(\tilde{s}_{t+h})] \\ &= \mathbb{E}[\tilde{s}_t \tilde{s}_{t+h}]. \end{aligned} \tag{27}$$

Notice, $\tilde{s}_t$ and $\tilde{s}_{t+h}$ are i.i.d. and, hence, independent. Thus,

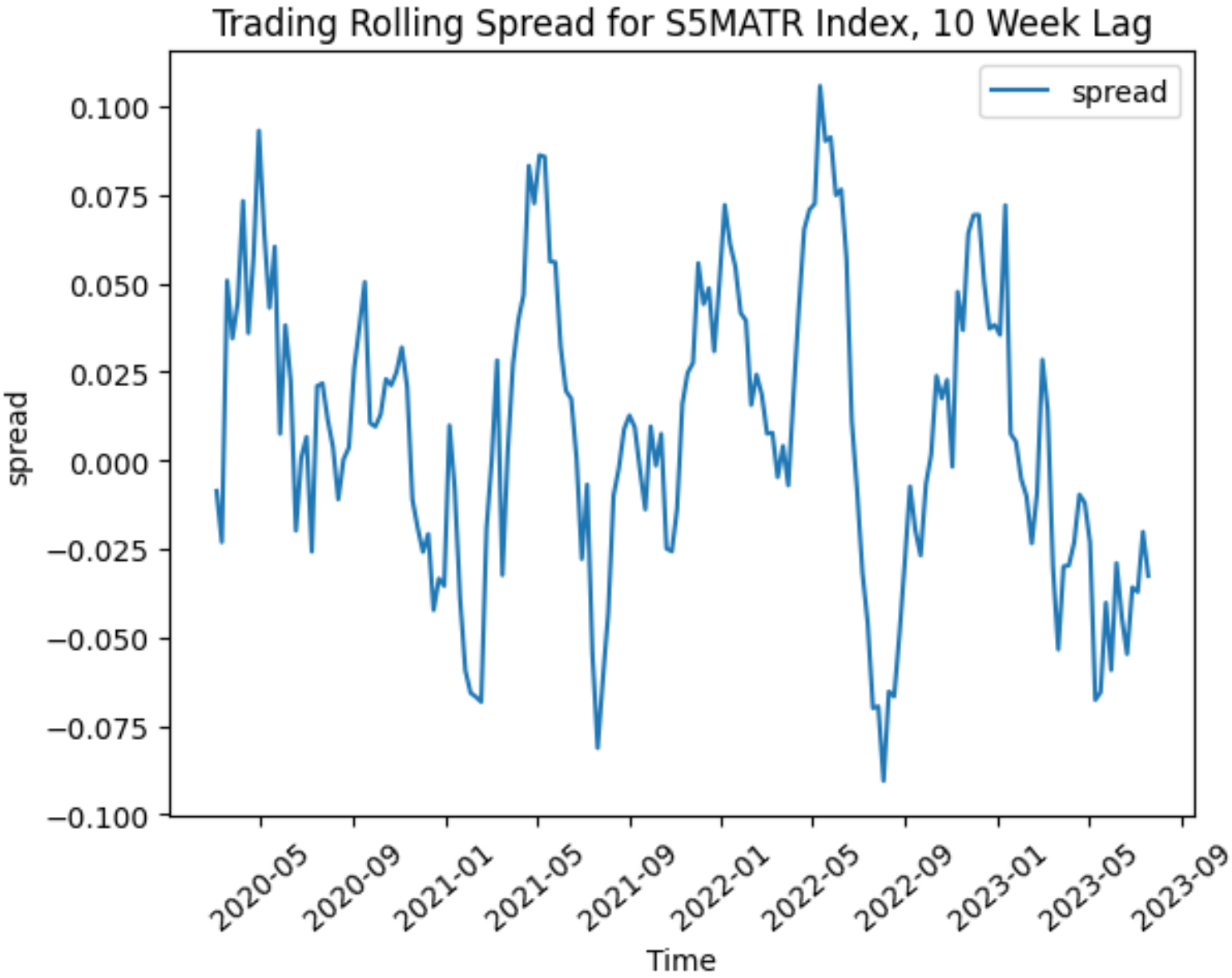
$$Cov(\tilde{s}_t, \tilde{s}_{t+h}) = \mathbb{E}[\tilde{s}_t]\mathbb{E}[\tilde{s}_{t+h}] = 0, \tag{28}$$

which concludes the proof. □

Refined In-Sample Spread Visualized



Refined Out-of-Sample Spread Visualized



z-score

- The z-score is, thus:

$$z_t = \frac{\tilde{s}_t - \mu_t}{\sigma_t}. \tag{29}$$

where μ_t and σ_t^2 is the rolling mean and variance of $\tilde{s}_t$, respectively.

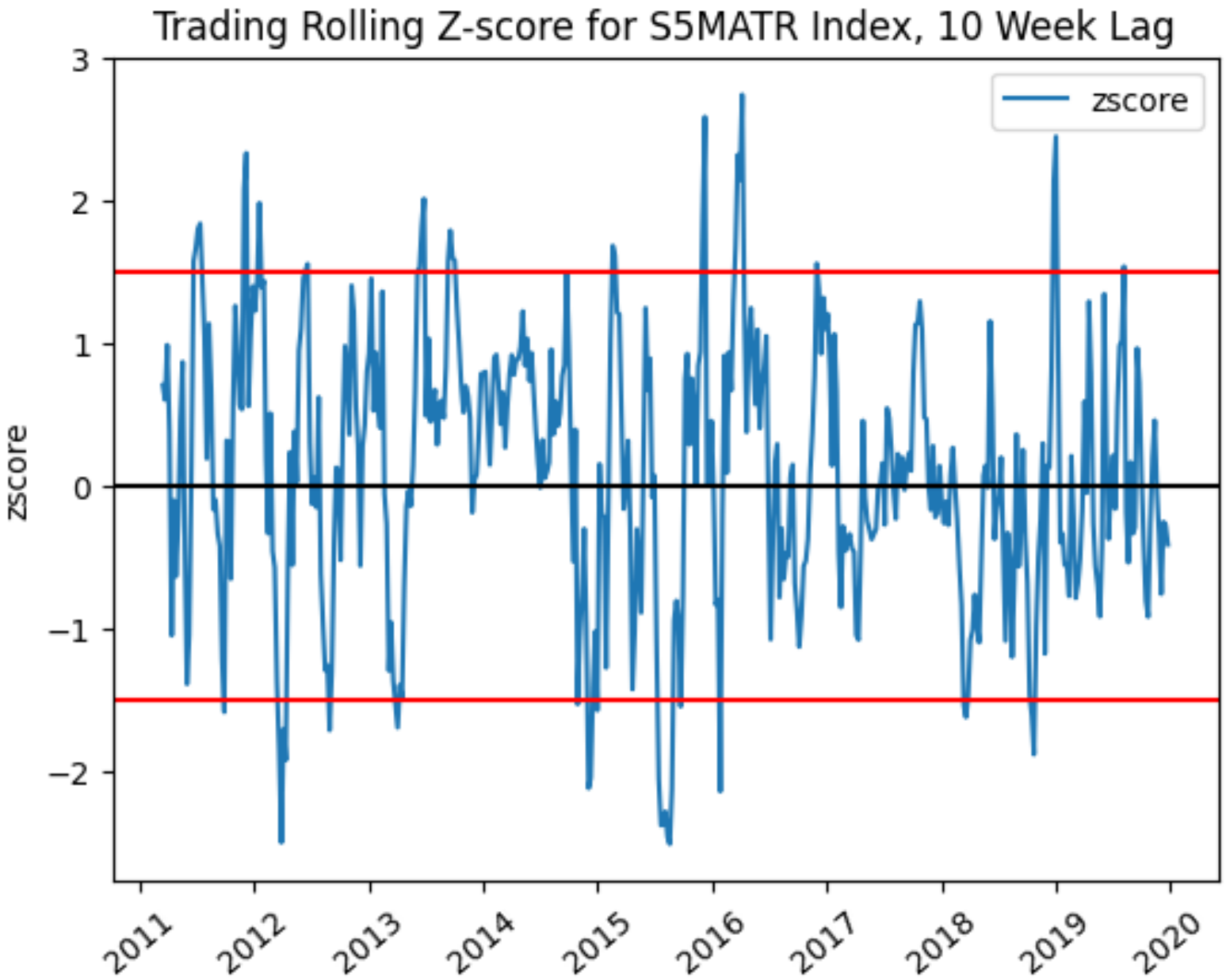
- That is:

$$\mu_t = \frac{1}{t}\sum_{i=1}^{t} \tilde{s}_i \tag{30}$$

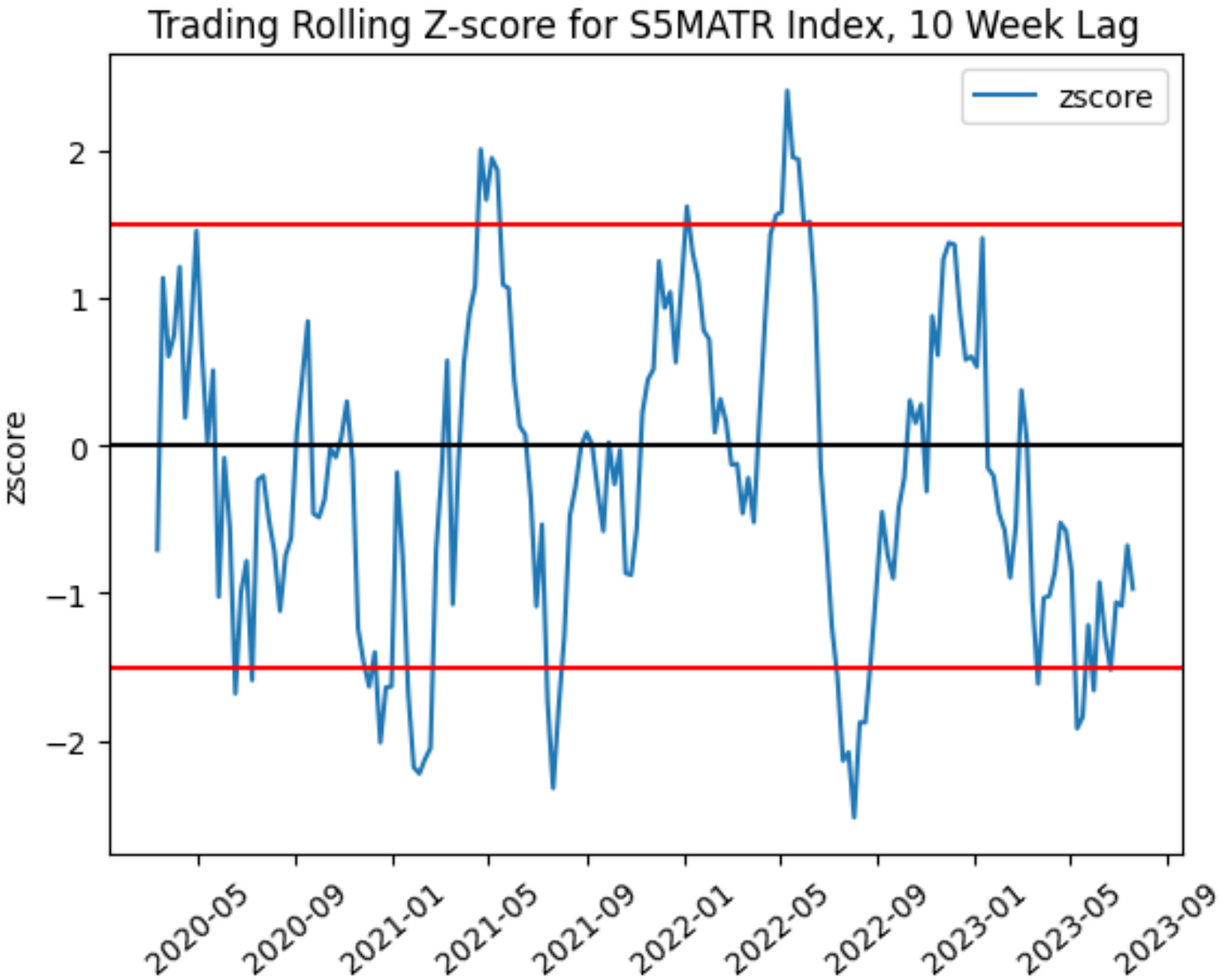
and

$$\sigma_t^2 = \frac{1}{t}\sum_{i=1}^{t} (\tilde{s}_i - \mu_t)^2. \tag{31}$$

In-sample z-score for Materials Sector



Out-of-sample z-score for Materials Sector



Stationarity

- While using a rolling sum makes the process technically non-stationary, it is weakly-stationary.
- Empirically, using a window of $N = 10$ weeks yields the best performance, in contrast to $N = 1$ week which produces a high β strategy.

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Defining the Trading Strategy

- From the characterization of $\tilde{s}_t$, it follows that z_t is weakly stationary and, hence, mean-reverting.
- If $z_t > +\alpha$, this means y_t is **overpriced** relative to our macroeconomic model; thus, we expect that it will mean-revert to a lower value. Therefore, we should short the asset.
- If $z_t < -\alpha$, we should long the asset, as it is **underpriced** relative to our macroeconomic model.
- Use $\alpha = 1.5$.

Choosing α

- While we chose $\alpha = 1.5$ for all sectors, we can do this more strategically.
- Let $z_t^- = \min(0, z_t)$ and $z_t^+ = \max(0, z_t)$ be the negative and positive part of z_t, respectively.
- Define $u_t := z_t^+ - \alpha$ and $v_t = -z_t^- - \alpha$.
- Then suppose we want the area of the parts above α and below $-\alpha$ to be η percent of the total signed area:

$$\begin{aligned}
\int_0^T u_t + v_t dt &= \eta \int_0^T z_t^+ - z_t^- dt \\
\int_0^T z_t^+ - z_t^- - 2\alpha dt &= \int_0^T z_t^+ - z_t^- dt \\
(1-\eta) \int_0^T z_t^+ - z_t^- dt &= 2\alpha T \\
\alpha &= \frac{1-\eta}{2T} \int_0^T z_t^+ - z_t^- dt
\end{aligned} \tag{32}$$

In-sample MAD Performance Metric

- We would like to identify the top-k performing sector models in-sample, according to the following 'goodness' of fit MAD metric:

$$g = w_1 \mathrm{AUC}_{\mathrm{norm}} + w_2 \mathrm{q}_{\mathrm{norm}} \tag{33}$$

  where $\mathrm{AUC}_{\mathrm{norm}} = \theta(\frac{AUC - \mu_{\mathrm{AUC}}}{\sigma_{\mathrm{AUC}}})$ is the normalized signed area-under-the-curve of the z-score z_t and $\mathrm{q}_{\mathrm{norm}} = \theta(\frac{q - \mu_q}{\sigma_q})$ is the normalized inverse p-value for a ADF stationarity test of z_t.
- Here, $\theta : \mathbb{R} \to [0, 1]$ is a logistic/sigmoid function.
- This metric accounts for (1) absolute divergence from the mean of zero and (2) stationarity of the z-score.

Top-5 Performing Models

- Top-5 performing sector MAD models: S5FINL Index, S5INDU Index, S5MATR Index, S5CONS Index, S5UTIL Index
- One could use relative returns as a means of quantifying MAD performance, but we would have to use a cross-validation set.

Statistical Significance of Signals

Null hypothesis: When fitting a MAD model to a given sector's weekly returns, we will analyze when that sector is +/- 1.5 stdev away from its determined MAD model, and we will see that there is no statistically significant difference in 1w, 2w, 3w, 1m, 2m, 3m, 6m, 9m, 12m relative returns of that sector to SPX following that +/- 1.5 stdev signal compared to the entire distribution of relative returns of that sector to SPX.

Return Distribution for Long-short

- We would like to characterize the statistical significance of the distribution of returns for a sector subject to a strategy where we enter a long/short position using each signal trigger.
- For it to be a reasonable test, the sample size must be relatively large.

Return Distribution for Long-short

Sector	Strategy	Duration	Hit Rate (%)	Median	Mean	Skew	One Sample T-Test	Number of Signals
S5FINL Index	Long	3m	0.00	-4.76	-5.07	-1.14	A	9
	Long	9m	22.22	-3.27	-2.36	-0.31	A	9
	Long	12m	11.11	-3.54	-3.25	0.66	A	9
	Short	3m	63.16	-0.78	0.87	0.31	A	19
	Short	9m	42.11	4.26	4.33	-0.04	A	19
	Short	12m	5.26	3.27	7.92	-0.17	A	19
S5INDU Index	Long	3m	100.00	6.04	6.28	0.09	R	4
	Long	9m	50.00	-0.34	2.49	1.06	A	4
	Long	12m	25.00	-0.85	3.07	0.98	A	4
	Short	3m	46.67	0.53	0.82	0.44	A	15
	Short	9m	73.33	-5.38	-2.28	0.84	A	15
	Short	12m	73.33	-5.30	-0.94	1.34	A	15
S5MATR Index	Long	3m	64.00	0.92	1.49	0.50	A	25
	Long	9m	44.00	-0.93	0.08	0.06	A	25
	Long	12m	36.00	-3.08	-0.30	1.50	A	25
	Short	3m	90.91	-7.26	-6.78	1.52	R	11
	Short	9m	54.55	-0.68	-1.56	-0.39	A	11
	Short	12m	90.91	-3.08	-2.94	1.42	R	11
S5CONS Index	Long	3m	33.33	-1.39	-1.07	0.32	A	15
	Long	9m	0.00	-7.25	-6.57	0.15	A	15
	Long	12m	20.00	-3.66	-4.19	0.09	A	15
	Short	3m	26.67	4.96	4.43	-0.10	A	15
	Short	9m	13.33	9.02	6.04	-1.51	A	15
	Short	12m	13.33	11.33	7.09	-1.30	A	15
S5UTIL Index	Long	3m	44.44	-5.12	-2.56	-0.20	A	9
	Long	9m	0.00	-17.06	-17.50	-0.37	A	9
	Long	12m	0.00	-19.68	-19.28	0.42	A	9
	Short	3m	44.44	0.19	-1.66	-0.42	A	9
	Short	9m	33.33	5.40	0.01	-0.77	A	9
	Short	12m	77.78	-3.14	-6.84	-0.02	A	9

Distribution of Relative Returns

- We have a maximum of 25 long signals for S5MATR and a maximum of 15 short signals for S5CONS.
- Note: this is not a very reasonable sample size to hypothesis test, but one can try to bootstrap inferences.

Distribution of Relative Long Returns for S5MATR Index

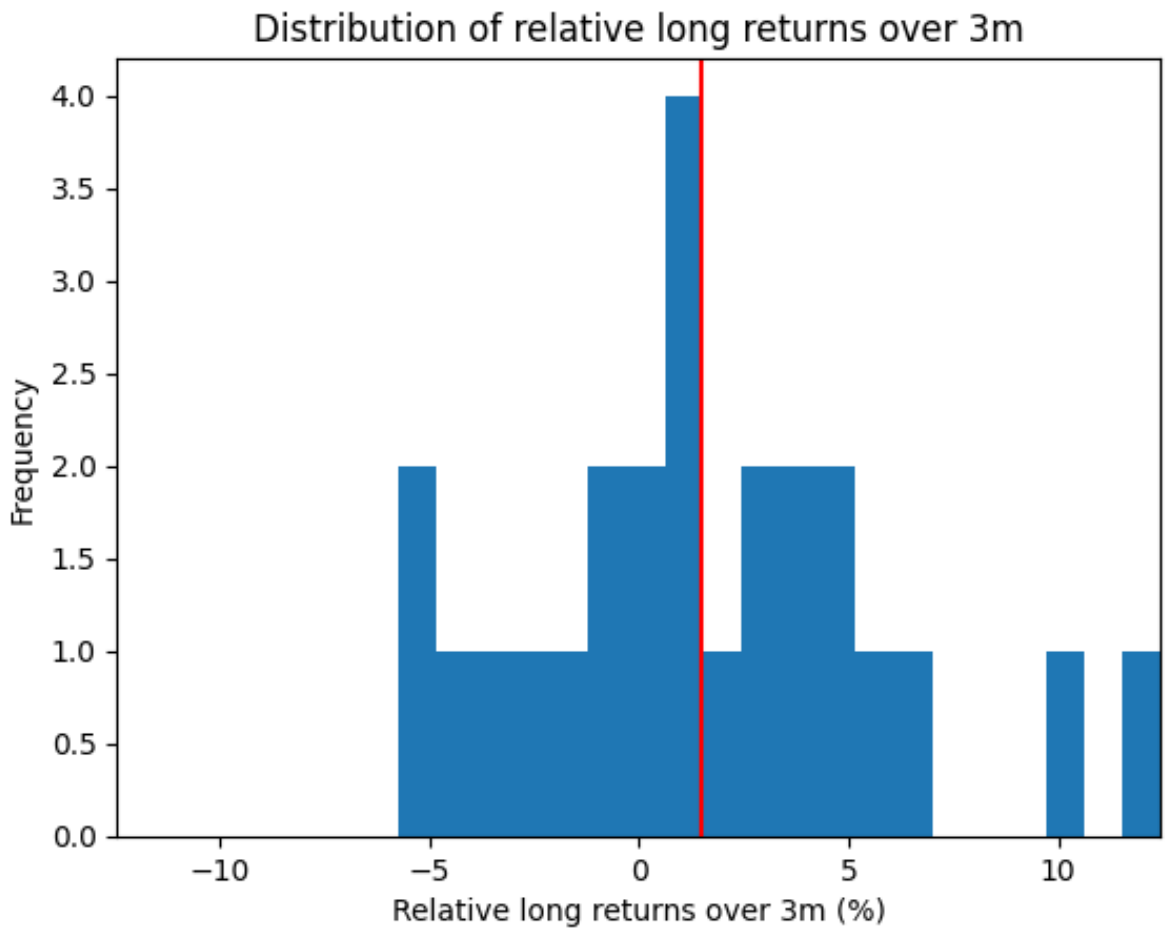


Figure: Distribution of long returns

Distribution of Relative Short Returns for S5FINL Index

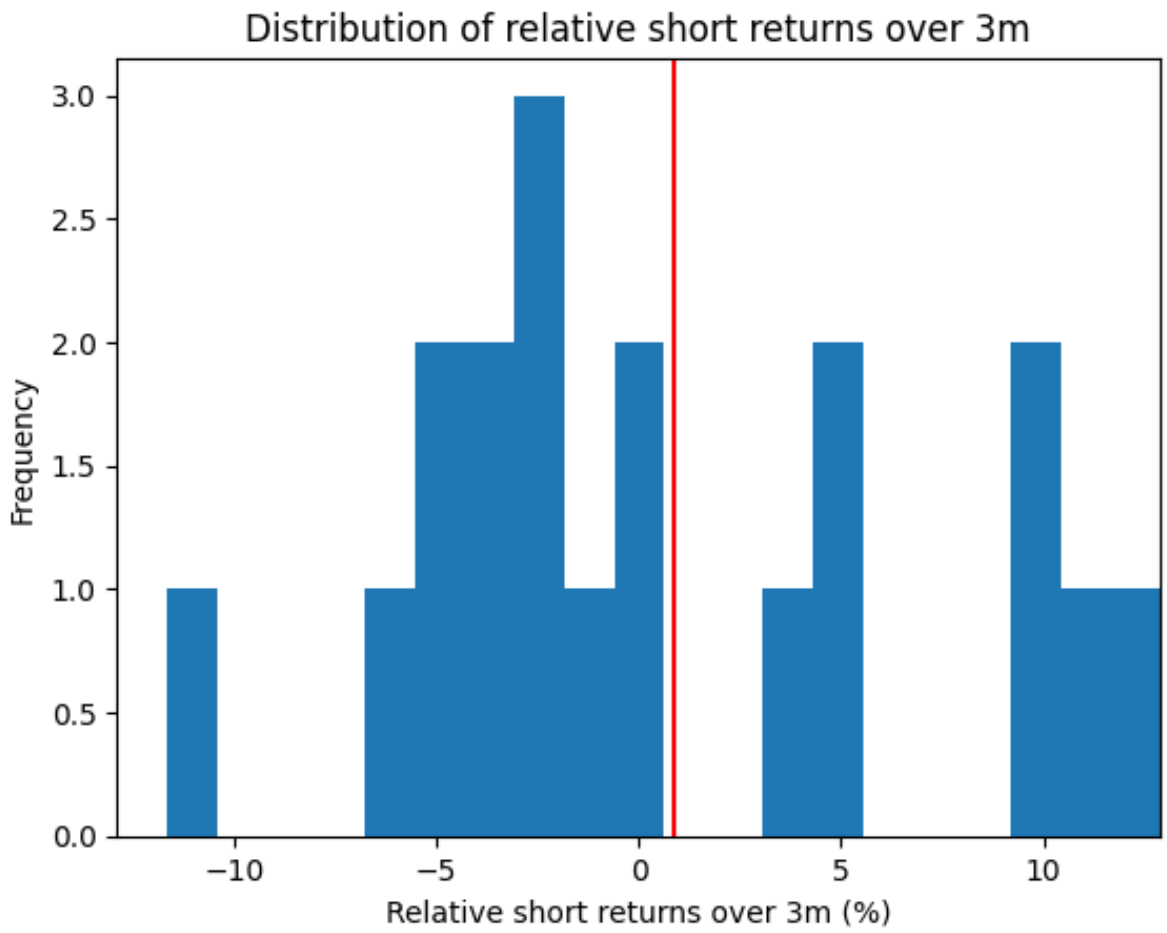


Figure: Distribution of short returns

Upshot

- Generally, the more signals, the stronger the statistical significance.
- Some sectors do not produce desirable returns; however, this simple analysis does not follow the exit trading rules that we will use.

Out-of-sample Portfolio Construction

- We define a counting process

$$X_t(s) := \sup\{0 \leq k \leq t | z_t(s) > +\alpha, \ldots, z_{t-k}(s) > +\alpha\} \tag{34}$$

as the number of consecutive short signals for sector s.

- Similarly, we define

$$Y_t(s) := \sup\{0 \leq k \leq t | z_t(s) < -\alpha, \ldots, z_{t-k}(s) < -\alpha\} \tag{35}$$

as the number of consecutive long signals for sector s.

Out-of-sample Portfolio Construction

- We construct a portfolio holding the top-5 performing sector indices: S5FINL Index, S5INDU Index, S5MATR Index, S5CONS Index, S5UTIL.
- We prorate the portfolio with weights according to the relative number of consecutive long (short) signals.
- This is a *momentum-based weighting*, whereby more consecutive long (short) signals means we should be more aggressive with our position.
- According to our discussion of statistical significance of signals, this is intuitive.

Out-of-sample Portfolio Construction

- Mathematically, we compute the long (short) portfolio weight of sector s at time t as:

$$w_t(s) = \frac{Y_t(s)}{\sum_r Y_t(r)}. \tag{36}$$

- Whenever there are no long/short signals, we allocate our entire portfolio into SPX.

Portfolio Simulation & Statistics

We deploy this long-short strategy over the out-of-sample period from January 2020 to August 2023.

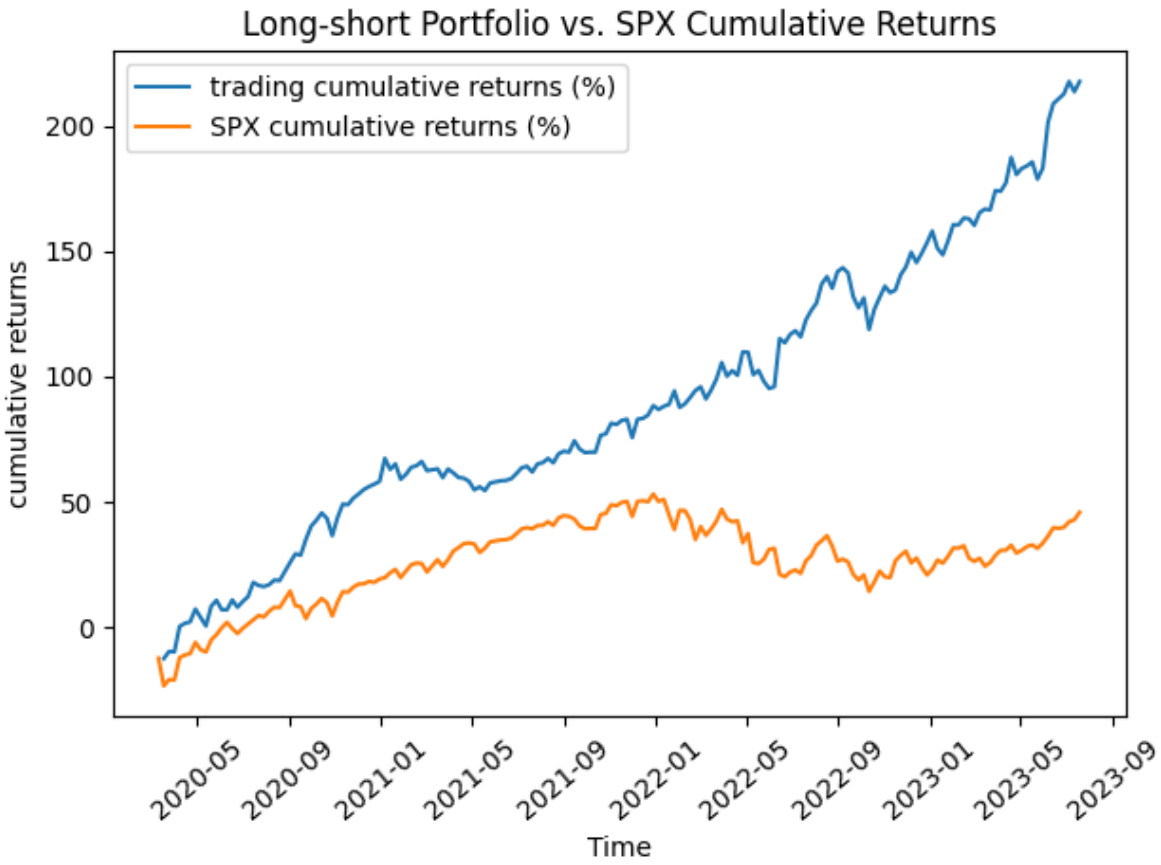


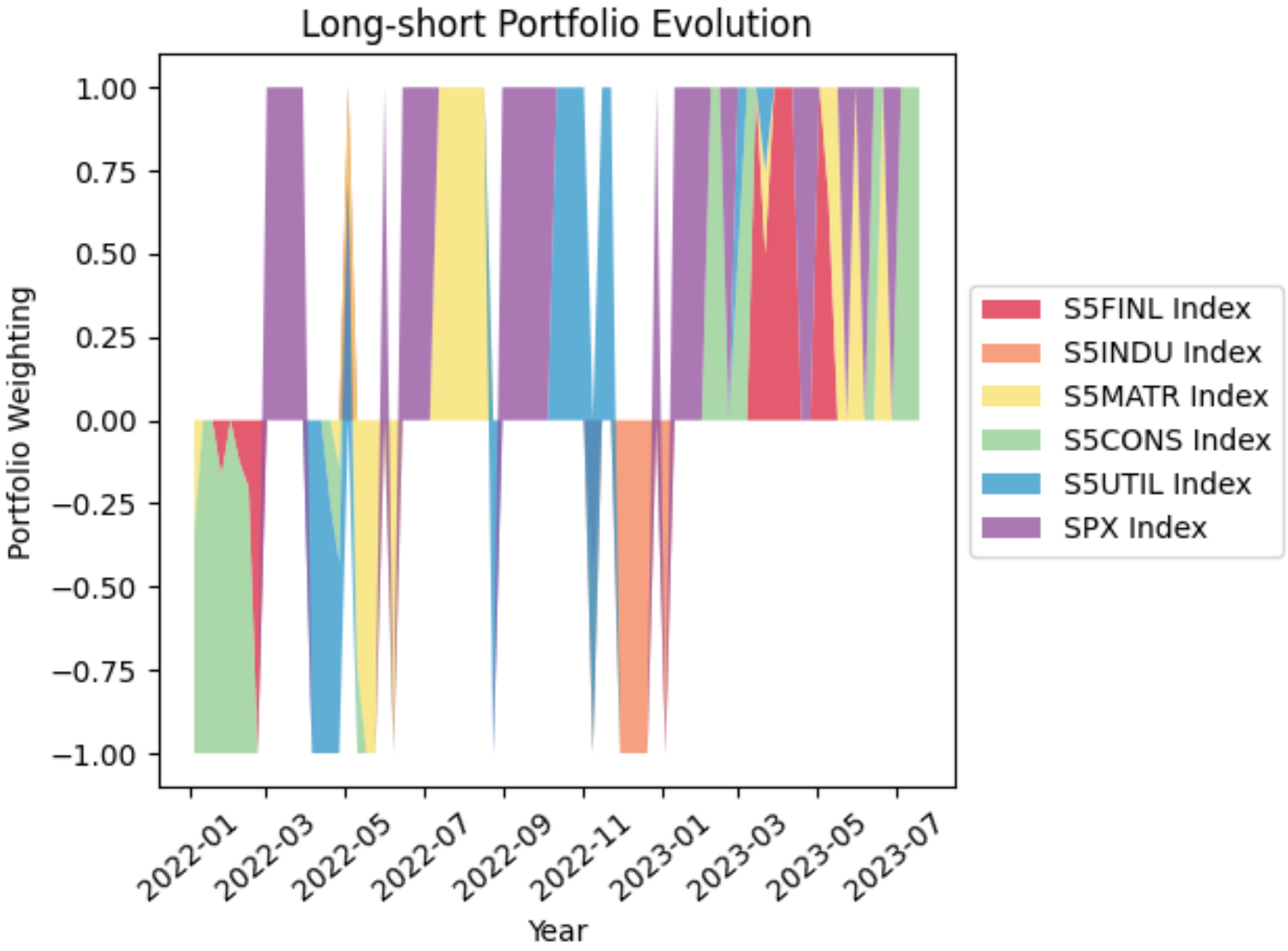
Figure: Cumulative Trading Returns for Portfolio Relative to SPX

Portfolio Statistics

	Total Return	APR	Sharpe	Max DD	Max DDD
S5FINL Index	0.548582	0.137934	0.632139	-0.269536	79.0
S5INDU Index	0.695996	0.168919	0.783509	-0.202770	85.0
S5MATR Index	0.747757	0.179348	0.791709	-0.242066	81.0
S5CONS Index	0.324150	0.086493	0.648824	-0.162876	65.0
S5UTIL Index	0.097931	0.027988	0.238355	-0.203595	48.0
Buy and Hold	0.476883	0.122106	0.699877	-0.197951	79.0
Algo (out-of-sample)	2.632638	0.467121	2.312357	-0.101219	32.0

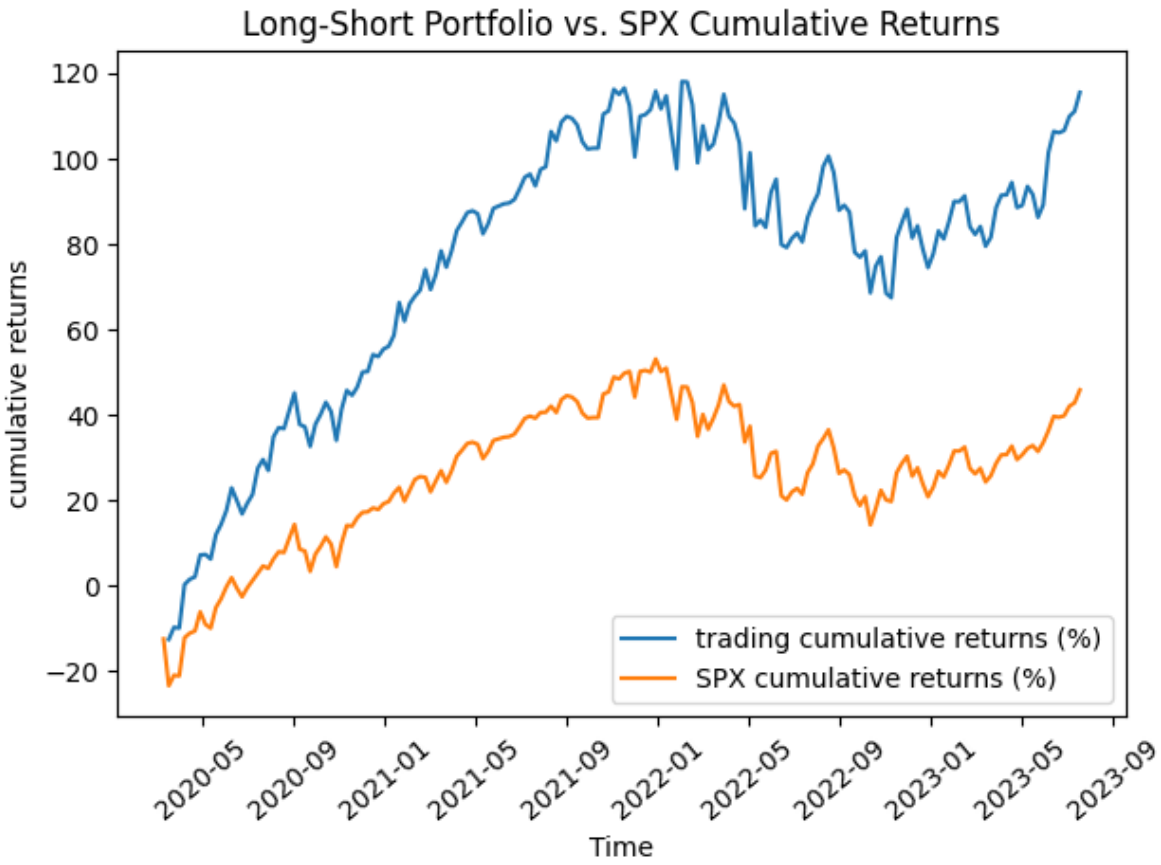
Table: Performance Metrics for Long-Short Strategy

Long-Short Portfolio Evolution



Long-only MAD Strategy

We also outperform relative to SPX using a long-only strategy from January 2020 to August 2023, albeit a higher β strategy.



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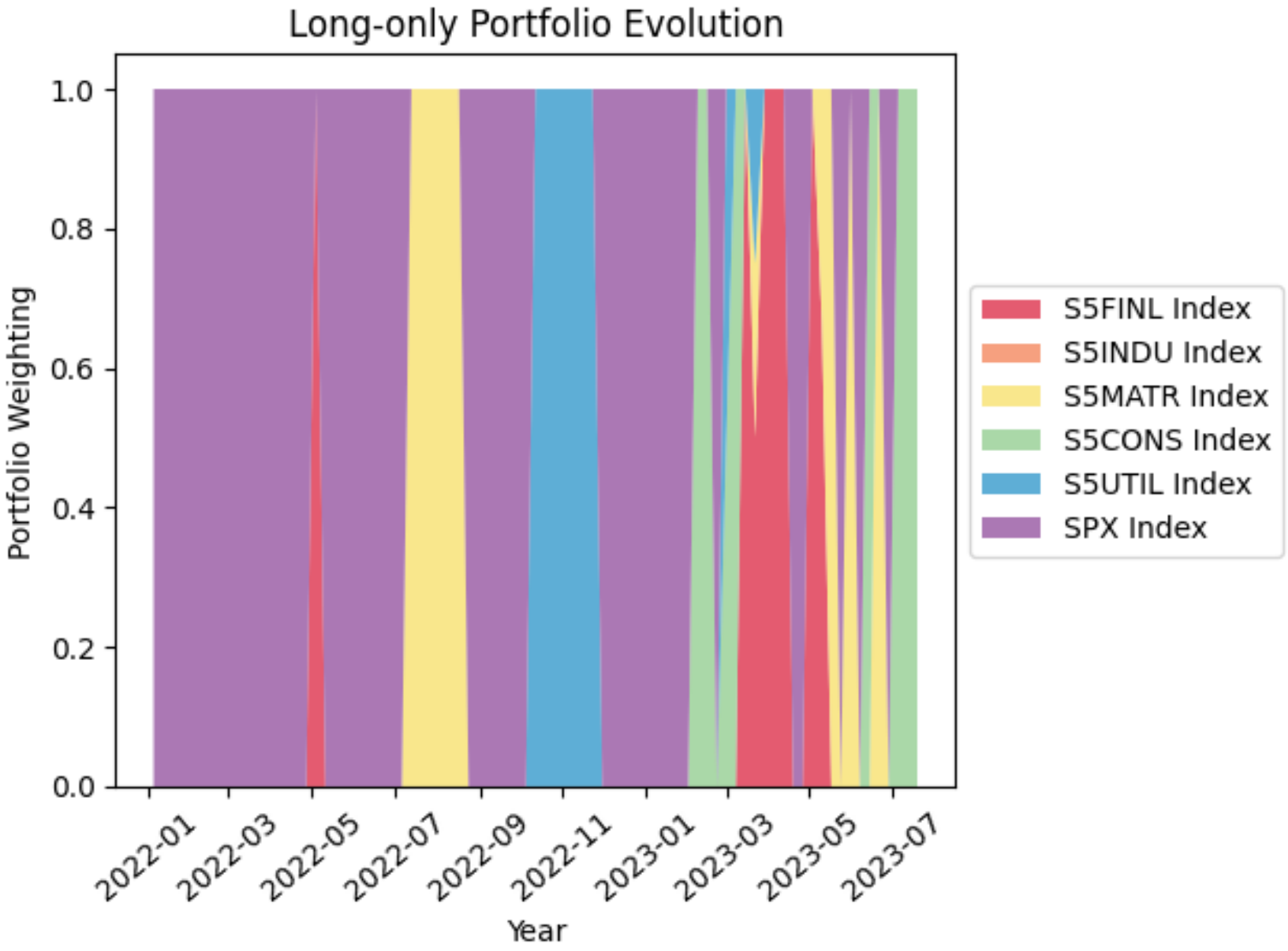
S5UTIL Index	0.097931	0.027988	0.238355	-0.203595	48.0
Buy and hold	0.476883	0.122106	0.699877	-0.197951	79.0
Algo (out-of-sample)	1.377279	0.293450	1.430722	-0.246549	74.0

Table: Performance Metrics for Long-only Strategy

Long-only Portfolio Evolution



Improvements

- We select the top-n performing MAD models using a metric that's hand-crafted to emphasize stationarity and divergence.
- While a good measure, this does not guarantee out-of-sample stationarity.
- We should watch for breaks in stationarity during trading; if this occurs, we re-train all models and repeat top model selection.
- One approach is using a *Kalman Filter* to dynamically track a hedge ratio vector (between multiple assets in the portfolio) to keep the spread stationary.
- Do a finer hyperparameter search over the spread lag parameter N and the signal threshold α.
- PCA for all sectors.
- Proper risk management.

Experience

- I really enjoyed my time here this summer and the opportunity to dig into interesting research ideas.
- I especially appreciated the flexibility I had in terms of exploring research directions and the types of projects I worked on.
- We constructed and tested hypotheses with a very statistically-sound focus.
- I kept my quantitative skills sharp, as well as refining and learning more financial principles from Alex and Bobby.

Next for Me

- Trading at quantitative hedge fund, explore interests in tech + finance.
- Complete masters in statistics.
- Goal: do math with real-world applications.
- I sometimes blog here: `https://sergiogcharles.github.io/`
- A special thank you to Alex and Bobby!
- Thank you to the ALPS team for having me this summer and everyone who has helped me throughout this internship!

Thank you!

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